

Computer Vision Proposal

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Idea: A macOS app that uses the webcam to detect hand gestures and switches desktops (Spaces) in response. Swipe left = previous desktop, swipe right = next desktop, with a couple more gestures if we have time (open palm to trigger Mission Control, fist to show desktop).

How: The pipeline has three parts. First, we extract hand pose information from the webcam feed using MediaPipe Hands, which gives us a stream of hand landmarks we can work with instead of raw pixels. This lets us focus on the gesture recognition problem itself. Second, we collect our own small dataset by each recording examples of every gesture under different lighting and angles, and train a lightweight classifier to recognize gestures over short time windows. We'll also implement a simpler rule-based baseline (something like "if the hand moves fast enough in one direction, call it a swipe") so we can compare a learned approach against a hand-coded one and talk about the tradeoffs. Third, we'll wire the output into macOS so detected gestures trigger the right keystrokes, with some logic to prevent the same gesture from firing repeatedly.

Tools: Python, OpenCV, MediaPipe, PyTorch, pynput.